INSTRUCTOR KEY AND RUN OF SHOW

The Case of the Silent Mineral: Raymond "Ray" Kessler, hypermagnesemia with AKI on CKD 4 (2-hour group case study)

not for students

HOW TO USE THIS KEY

- Print one student packet per squad (2 to 3 students). Open the EHR with Ray (Pre-Built Patients > Case Study Patients > Raymond Kessler) on the instructor screen, clock **Tue 0700**. Students use their own devices with the same patient and synced time.
- Open `monitor.html` on the projector (it follows the clock). Each checkpoint you press **4h**, read the Situation, and the monitor changes to that checkpoint. The Bedside Nurse reads the vitals from the monitor and charts them.
- Squads rotate three jobs: A Chart Detective, B Lab Interpreter, C Bedside Nurse. Two-person squads: Desk (A+B) and Bedside (C), swapping each checkpoint; skip stretch boxes.
- The packet is 27 pages: cover, rubric, patient card, role cards, handoff, six checkpoint sections (3 pages each; the 1500 section is 4), and 3 Case Closed pages. Students must not read ahead.
- This key lists a model answer for every box: the answer, why (tied to the chart), what to look for, common misses. Full credit means complete and thoughtful, not word-for-word.

THREE THINGS THAT GO WRONG

1. Squads that finish early read ahead: say "stay on your page until I press 4h".

2. Nobody asks Eleanor about OTC products at 0700: release her line at about 0:16 (see 0700 row).

3. A student gives a magnesium product or a sedative in the EHR: pause, debrief it right there (see INSTRUCTOR_GUIDE.md section 9), then reset the monitor.

RUN OF SHOW: ABOUT 2 HOURS, MINUTE BY MINUTE

press 4h at the end of each segment

Start	Segment	What the instructor does (read, portray, circulate)	If running behind
0:00 10 min	OPENING (EHR clock: Tue 0700, monitor at the 0700 row)	0:00-0:03 Hand out one packet per squad. Squads of 2-3 write names and starting jobs on the cover; point at the rubric (page 2) and say "complete and thoughtful = full credit". 0:03-0:05 Show the roles page and the rotation chart. Tell them: "Do not turn a page past your checkpoint until I press 4h." 0:05-0:10 Read the ED handoff (script in the Opening section below). Students fill the S-B-A-R listening grid and read back. Circulate: check the read-back (name, DOB, allergy). None yet. Monitor shows the 0700 set (HR 52, BP 96/56, RR 14, SpO2 94%).	If behind: skip the "asked one clarifying question" line and the rhythm decoder explanation.
0:10 15 min	CHECKPOINT 1: Tuesday 0700 (First Look)	Clock Tue 0700. Read Situation (page 1). Monitor: 0700 row (INSTRUCTOR_GUIDE.md section 4, Checkpoint 1 (B)). Portray Ray per section 4 Checkpoint 1 (C); Eleanor answers only if asked about OTC products, laxatives, antacids or supplements, or asks "should I show you the bag?" at about 0:16. Circulate at 0:14 (is the Bedside Nurse really reading the monitor and charting in Flowsheets?) and 0:17 (has anyone asked about the bag? if not, release Eleanor's line). Listen at the huddle for ABC first. Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 1B2 (rhythm decoder), then 1A2(b) (the hunt). Do not skip vitals, the Mg-o-meter, priorities or the SBAR warm-up.
0:25 15 min	CHECKPOINT 2: Tuesday 1100 (Code Yellow)	Press 4h (clock Tue 1100). Read Situation. Monitor: 1100 row (section 4 Checkpoint 2 (B)): HR 44, BP 84/48, RR 10 shallow, SpO2 91% room air (94% on 3 L if they apply oxygen). Portray the mumbling patient with absent reflexes (Checkpoint 2 (C)). Circulate at 0:29: is a "call for help" decision being made before 0:33? If nobody says "rapid response", ask as the charge nurse: "What is his respiratory rate? What would the criteria say?" At the huddle, play the rapid response doctor: "Tell me what you need." (SBAR #1 in the guide). Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 2B3 (rhythm), then 2A3 ("what is missing"). Never skip the SBAR.
0:40 15 min	CHECKPOINT 3: Tuesday 1500 (The Big Decision)	Press 4h (clock Tue 1500). Read Situation. Monitor: 1500 row (section 4 Checkpoint 3 (B)): HR 46, BP 88/50, RR 9, SpO2 92% on 4 L, EtCO2 52, wide QRS. Crackles at the bases; dialysis RN at the bedside (Checkpoint 3 (C)). This section is 4 pages (the machine page is page 3). Circulate at 0:44 (are they using the ICU outreach note numbers?) and 0:50 (huddle: airway first?). Play ICU outreach for the SBAR (#3 in the guide). Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 3B4 and the diagram page (page 3 except the Mg-o-meter), then 3A3. Do not skip the dialysis readiness check or the SBAR.
0:55 15 min	CHECKPOINT 4: Tuesday 1900 (The Quiet After the Storm)	Press 4h (clock Tue 1900). Read Situation. Monitor: 1900 row (HR 66, BP 104/60, RR 16, SpO2 96% room air). When the Bedside Nurse checks glucose say "54"; after the first 15 g say 68, after the second 84 (Checkpoint 4 (B)). Patient: "shaky, sweating, my mouth and hands tingle". Circulate at 0:59: orange juice? swallow screen first? Listen for the juice choice at the huddle. Play the hospitalist for SBAR #4. Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 4B3, then 4A3. Do not skip the glucose steps.
1:10 15 min	CHECKPOINT 5: Tuesday night 2300 (The Night Shift Riddle)	Press 4h (clock Tue 2300). Read Situation. Monitor: 2300 row (HR 62, BP 112/64, RR 16, SpO2 96%, POC 118). Portray confusion: "I have got to get to the shop. Get these things off me! Where is Eleanor?" (Checkpoint 5 (C)). Circulate at 1:14: is anyone reaching for the PRN sleep medicine or restraints? Prompt: "Read the drug class. What does it do at a magnesium of 5.6?" Play the night hospitalist: "Is the diphenhydramine order active? Do not give it." (SBAR #5). Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 5B3, then 5A3.

1:25 15 min	CHECKPOINT 6: Wednesday 0300 (The Boomerang)	Press 4h (clock Wed 0300). Read Situation. Monitor: 0300 row (HR 56, BP 104/60, RR 12, SpO2 95%, PR 0.24 s, POC 126). Ray: "My belly is so full. Can I take something? At home I take the pink milk." (Checkpoint 6 (C)). Circulate at 1:29: if the milk of magnesia step is not found, prompt "Read every ingredient of the protocol. Which step has magnesium?" Play the provider: "Good catch: do NOT give it. Discontinue the bowel protocol. Bisacodyl 10 mg suppository. Get a KUB. No Fleet or magnesium enemas." (SBAR #6). Timeline inside the 15 min: :00-:02 read Situation and set monitor; :02-:10 role boxes; :10-:14 team huddle; :14-:15 collect and press 4h.	If behind: skip 6A3, then 6B3.
1:40 15 min	CASE CLOSED: Wed 0700 (final check and debrief huddle)	Press 4h one last time (clock Wed 0700). Read the Situation on the Case Closed page. Monitor: 0700 row (HR 68, BP 124/70, RR 16, SpO2 96%, POC 112; weight 90.0 kg). Ray: "I feel so much better. I want a cup of coffee. I did not know all those pills were a problem." Eleanor: "What do I do with all those bottles at home?" 1:40-1:45 final check (vitals, labs, Mg-o-meter whole story). 1:45-1:55 debrief huddle (questions below). Teach-back and individual reflections: do them in the last 3 minutes, or send them home to finish and turn in with the paperclip next class.	If behind: shorten the timeline to four stops; teach-back as homework.

Monitor pointer for every row: INSTRUCTOR_GUIDE.md section 4 lists the exact monitor table (B), bedside script (C), chart contents (D), expected actions (E) and prompts (F) for each checkpoint. Section 5 is the release schedule if you use 1h or Next event instead of 4h. Section 6 holds the model SBARs.

OPENING: HANDOFF AND ROLES

Tue 0655-0700

Read aloud as the ED nurse (K. Ortiz, RN):

S: "Raymond Kessler, 74-year-old man, arriving at 5E-14 with critical hypermagnesemia (Mg 7.8; normal 1.7 to 2.2), AKI on CKD 4 (creatinine 4.9), a slow heart rate and low blood pressure. Full Code. Allergy: penicillin, a rash."

B: "Four days of vomiting and poor intake, then hard to wake. At home: lisinopril, furosemide and ibuprofen. His wife says he takes stomach medicine and a magnesium pill. A bag of bottles was delivered at 0640 and has NOT been reviewed. Last stool Friday. Home meds are on hold. In the ED: ondansetron at 0510, calcium gluconate 1 g at 0555, a 500 mL saline bolus 0605 to 0635, and a Foley at 0610."

A: "At 0700: temperature 97.6, heart rate 52 sinus bradycardia with first-degree AV block, BP 96/56, RR 14, SpO2 94% on room air, pain 2 out of 10. Drowsy but wakes to voice, slurred speech, reflexes 1+, weak grip, flushed, dry mouth, abdomen a little distended. NPO. Lines: 20 gauge left forearm and 18 gauge right antecubital. Foley 16 French, 130 mL out."

R: "Second saline bolus 500 mL is due at 0715: listen to the lungs before and after. Then 100 mL/h. Furosemide 80 mg IV at 0915. Magnesium and BMP at 0900 then every 4 hours. Pacing pads available. Calcium gluconate as needed. Please review the bag of bottles with his wife. Hourly urine output."

H1 Situation

Who is this and what is wrong right now?

Model answer: Raymond "Ray" Kessler, 74 M, room 5E-14, arriving from the ED with critical hypermagnesemia (Mg 7.8; normal 1.7-2.2), AKI on CKD 4 (creatinine 4.9), a slow heart rate and low blood pressure. Full Code. Allergy: penicillin (rash).

Why (tied to the chart): Straight from the 0655 ED handoff note.

Look for: Name, age, problem, a number and the allergy.

Common misses: Leaving out the allergy or code status.

H2 Background

How did he get here? What did the ED do, and when?

Model answer: 4 days of vomiting and poor intake, then hard to wake. Home lisinopril, furosemide and ibuprofen. Wife says he takes "stomach medicine" and a "magnesium pill"; a bag of bottles (delivered 0640) has NOT been reviewed. Last stool Friday. In the ED: ondansetron 0510, calcium gluconate 1 g 0555, NS 500 mL 0605-0635, Foley 0610.

Why (tied to the chart): ED handoff 0655 and ED provider note.

Look for: Cause, a medication clue, and the three ED treatments with times.

Common misses: Missing the unreviewed bag.

H3 Assessment (0700)

What did the ED nurse see at 0700?

Model answer: T 97.6, HR 52 sinus bradycardia with first-degree AV block, BP 96/56, RR 14, SpO2 94% room air, pain 2/10. Drowsy but wakes to voice, slurred speech, reflexes 1+, weak grip, flushed, dry mucosa, abdomen mildly distended, NPO. Lines: 20 g left forearm and 18 g right antecubital; Foley 16 Fr (130 mL out).

Why (tied to the chart): ED handoff 0655.

Look for: Vitals, level of consciousness, reflexes.

Common misses: Writing only the vital signs.

H4 Recommendation: what is due next?

What is still to do? What worries you?

Model answer: Second NS 500 mL bolus due 0715 (lung check before and after), then 100 mL/h; furosemide 80 mg IV at 0915; Mg and BMP at 0900 then q4h; pacing pads available; calcium gluconate PRN; review the bag of bottles with the wife; hourly urine output.

Why (tied to the chart): ED handoff 0655.

Look for: Two scheduled items and the unreviewed bag.

Common misses: Missing the bag.

Not scored by itself: the handoff listening notes feed the Teamwork and Clinical reasoning impressions. The rhythm decoder (strips A, B, C) and the reflex scale are reference tools for the Lab Interpreter and the Bedside Nurse.

CHECKPOINT 1: FIRST LOOK

Tuesday 0700 | 3 student pages

Read-aloud Situation (also printed in the packet):

It is Tuesday 0700 on 5 East, room 5E-14. Ray Kessler, 74, came up from the ED about ten minutes ago. He has kidney disease, diabetes and a weak heart. He vomited for four days and then could not be woken this morning.

The ED found a very high magnesium (7.8; normal is 1.7 to 2.2) and a creatinine of 4.9. They gave calcium gluconate, a saline bolus and placed a Foley. His wife Eleanor sits by the bed holding a paper bag from home. She has not been asked about it yet.

Set the monitor to	Value
Temp (F)	97.6
HR	52
BP (MAP)	96/56 (69)
RR	14
SpO2 / O2	94% room air
EtCO2	not shown
Pain	2/10 low back
Rhythm	Sinus bradycardia with first-degree AV block (PR 0.26 s)
POC glucose	132

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness (RASS) and orientation	Drowsy, wakes to voice, falls asleep when you stop talking (RASS -1); oriented to person and place, not the date; slow, slurred speech
Breathing: rate for a FULL minute, depth, pattern	RR 14, regular, unlabored, normal depth
Reflexes: patellar and biceps (0 to 4+)	Patellar 1+ and biceps 1+ both sides (diminished; normal is 2+)
Strength: grip and lifting the legs	Lifts all four limbs against gravity; weak grip both hands (4-/5), tires quickly
Lungs (before any fluid!) and neck veins	Clear, diminished at both bases; neck veins flat at 30 degrees; no edema
Skin and mouth	Face and chest flushed and warm; dry mucous membranes; skin tents for 3 seconds over the sternum
Urine (Foley): color and amount	Dark amber, clear; 130 mL since the Foley went in at 0610 (ED block 0430-0700); the plan for 0700-0800 is 35 mL

Skip order if behind: 1B2, 1A2(b). Safety/SBAR point: warm-up only (not scored).

1C1 Your 20-second report

Say it out loud to the team, then write it: the three findings that worry you most, and why.

Model answer: Any three of: HR 52 with a long PR (first-degree AV block) and low BP 96/56; drowsy and slurred; reflexes only 1+ (weak) and weak grip; flushed, dry, tenting skin (dehydrated); SpO2 94% on room air.

Why (tied to the chart): Magnesium blocks nerve-to-muscle signals and slows the heart. Reflexes are the bedside early warning, breathing fails later. The vitals order says to call for RR under 12, HR under 45, SBP under 90, lost reflexes or any drop in alertness: Ray is close to several of them.

Look for: Full credit = three specific findings with a reason tied to magnesium or dehydration, not just "he is sleepy".

Common misses: Only listing vital signs and skipping the reflexes; saying "SpO2 is fine so he is fine".

1A1 The evidence bag

Eleanor opens the bag (the instructor tells you when: ask her about over-the-counter products!). For each bottle, read the label and fill in the table.

#	Active ingredient (from the label)	Mg? Y/N	Why it matters for Ray
1 Milk of Magnesia	magnesium hydroxide	Yes	Laxative taken 30 mL 3-4 times a day for 2 weeks; absorbed magnesium cannot be cleared
2 Antacid (Maalox type)	aluminum + magnesium hydroxide	Yes	Heartburn/nausea: after meals and bedtime, plus extra doses for 4 days

3 Magnesium oxide 400 mg	magnesium oxide	Yes	Supplement for leg cramps, twice a day for 3 weeks
4 Ibuprofen 200 mg	ibuprofen (NSAID)	No	Narrows kidney blood flow; 600 mg three times a day for 2 weeks: AKI risk, worse with an ACE inhibitor and dehydration
5 Lisinopril 20 mg	lisinopril (ACE inhibitor)	No	Lowers kidney filtering pressure when dry; raises potassium (5.4); lowers BP
6 Furosemide 20 mg	furosemide (loop diuretic)	No	Water pill: dehydrates him when he is already vomiting

Model answer: Three of six bottles contain magnesium (1, 2, 3); the other three hurt the kidneys (4, 5, 6).

Why (tied to the chart): Chart: pharmacy note 0750 and the four OTC stop orders; DESIGN: doses and durations. Eleanor will not volunteer them unless asked about OTC products, laxatives, antacids or supplements.

Look for: Full credit when all three magnesium products are identified with the ingredient word "magnesium", plus a kidney reason for any two others.

Common misses: Calling ibuprofen a magnesium product; missing that the antacid also contains magnesium.

1A2 MAR and orders sweep

(a) The saline bolus is due at 0715. What must your team do before it starts and after it ends? (Ray has a weak heart: HFpEF.) (b) DETECTIVE HUNT: read the Active orders all the way down, including standing "order set" protocols. Is anything there with magnesium, phosphate or a sedative? Write what you found and where, or "not found yet".

Model answer: (a) Listen to lungs, check SpO2 and neck veins BEFORE and AFTER; stop and call for crackles, SpO2 under 92%, new edema or weight gain; hourly urine. (b) The standing bowel protocol (senna, then bisacodyl suppository, then milk of magnesia 30 mL if no stool in 72 hours) ends in a magnesium product, and the admission order set also has diphenhydramine 25 mg PRN for sleep (sedating). Both sit right next to an active order that says NO magnesium products.

Why (tied to the chart): This is the hidden suspect of the case (ordered 0639). The hunt is practice: finding it now is a bonus, but nobody loses points for "not found yet" if they say where they looked. It returns at 2300 and 0300.

Look for: Credit (a) for lungs/SpO2 before and after AND a reason (heart failure). Credit (b) for looking in the orders and naming at least where they looked.

Common misses: Only reading the order titles. The magnesium is in the last step of the protocol.

1B1 Copy the labs

Copy the 10/20 clinic and ED 0440 values. Then write H (higher than normal) or L (lower).

Test	Oct 20 clinic	Tue 0440 (ED)	H or L?
Magnesium	2.4	7.8	H, H (7.8 is critical)
Potassium	4.8	5.4	normal, H
Creatinine	2.6	4.9	H, H
Calcium	9.0	8.1	normal, L

Model answer: See the table. eGFR fell from 25 to 12.

Why (tied to the chart): Practice reading the Lab Results grid and comparing to baseline. Baseline creatinine 2.6 is the CKD 4 starting point; 4.9 is acute kidney injury on top.

Look for: Accept the stated units or none. Score 1 for mostly right values and H/L.

Common misses: Copying the wrong column (the 0430 and 0450 columns hold only POC glucose, ECG and urine tests).

1B2 Rhythm decoder

The ED ECG (0430): HR 46, PR 248 ms, QRS 124 ms. At 0700 the monitor shows PR 0.26 s. Which strip on the Handoff page is closest, A, B or C? Finish: "Magnesium makes the heart…".

Model answer: B (long PR = first-degree AV block); QRS 124 ms is just over the 120 limit, so the QRS is starting to widen toward C. Magnesium slows the signal as it travels through the heart (blocks calcium channels), so the heart is slower and the PR longer.

Why (tied to the chart): Sets up the 1500 ECG (strip C). Normal PR is under 200 ms and QRS under 120 ms.

Look for: Accept B with a note about slowing conduction.

Common misses: Choosing C: the wide QRS and tall T come later.

1B3 Connect the dots

In your own words: why can Ray's magnesium climb so high? Use two numbers from your table.

Model answer: Kidneys are the exit door for magnesium. Ray's kidneys were already weak (CKD 4: eGFR 25, creatinine 2.6) and dropped further (eGFR 12, creatinine 4.9, BUN 84) with vomiting, dehydration and kidney-hurting medicines, so magnesium could not leave.

Why (tied to the chart): Chart: eGFR 25 to 12; creatinine 2.6 to 4.9; Mg 2.4 to 7.8.

Look for: Full credit = kidneys + two real numbers.

Common misses: Saying "he took too much" without the kidney link (a healthy kidney clears extra magnesium).

1M Mg-o-meter

New level (Tue 0440): _____ mg/dL Zone: _____

Model answer: X at 7.8 (Tue 0440), in the REFLEXES LOST band (7-10). Ray's reflexes are weak (1+) rather than gone: he is climbing toward that zone.

Why (tied to the chart): Bands overlap in real life; the exam matters as much as the number. 7.8 mg/dL = 6.4 mEq/L.

Look for: Accept zone 7-10 (or "just under reflexes lost, still weak"). Means: critical, still rising, heart and muscles at risk.

Common misses: Marking the clinic 2.4 again or putting the X between columns.

1T1 Top 3 priorities

List the top 3 priorities for Ray's first hour, in order, with a few words on why.

Model answer: 1) Airway, breathing, circulation and neuro checks: full-minute RR, SpO2, monitor on, HOB 30, suction at the bedside, NPO until a swallow screen, reflexes and LOC. 2) Find and stop every magnesium source: ask Eleanor about over-the-counter products, review the bag, tell the provider and pharmacy, make sure no order or protocol adds magnesium. 3) Fluids safely and kidneys: lung check before and after the 0715 bolus, hourly urine, hold the home medicines that hurt kidneys, heart or breathing.

Why: Order rationale: magnesium toxicity kills by stopping breathing and the heart; the cause must be removed; fluids can help or hurt (HFpEF).

Look for: Full credit = 3 sensible priorities in a defensible order with a reason for #1 (airway/breathing first). Accept calcium gluconate ready, pacing pads located, call parameters.

Common misses: Putting "teach the family" or "call the doctor" first, or leaving out airway/breathing.

1T2 One question for the provider

The ONE question we will ask the provider (make it specific to Ray):

Model answer: Examples: "Can nephrology see him now instead of routine, with Mg 7.8 and eGFR 12?" "Can you remove the magnesium step from the bowel protocol and tell me when to give calcium gluconate?" "What are my call numbers for his breathing and heart rate?"

Look for: A good question is specific, uses a number from the chart and asks for a decision or an order.

Common misses: Vague ("What should I do?").

1T3 SBAR

Practice call: tell the hospitalist what you found in the bag. (Use S-B-A-R.)

Model answer: S: "This is the RN for Raymond Kessler in 5E-14. His wife brought a bag of home products: milk of magnesia, a magnesium antacid, magnesium oxide, ibuprofen." B: "74 years old, CKD 4 with AKI (creatinine 4.9), Mg 7.8 on arrival, HR 52 with long PR, BP 96/56, reflexes 1+." A: "The magnesium load comes from these products and his kidneys cannot clear it." R: "I would like all of them discontinued, pharmacy to review, nephrology to see him, and the bowel protocol checked for magnesium."

Why: Warm-up SBAR: not scored in Safety/SBAR for this checkpoint (the 1100 to 0300 calls are).

Look for: Look for all four letters, patient ID, one number per letter and a clear request.

Common misses: Skipping the Recommendation.

1P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #1**: Kidneys are the exit door for magnesium. When the door is almost shut (eGFR 12), every swallow of a magnesium product adds to the pile. Write it on your case file: Suspect = the bag.

CHECKPOINT 2: CODE YELLOW

Tuesday 1100 | 3 student pages

Read-aloud Situation (also printed in the packet):

It is 1100. Since 0700 Ray has had another saline bolus, maintenance saline, and a water pill (furosemide) by IV at 0915. The pharmacist went through Eleanor's bag and the home products were stopped.

The 0900 labs are back: magnesium 8.7, up from 7.8. Eleanor stepped out to call their daughter. You walk in to check on Ray.

Set the monitor to	Value
Temp (F)	97.4
HR	44
BP (MAP)	84/48 (60)
RR	10, shallow
SpO2 / O2	91% room air
EtCO2	not shown
Pain	0/10
Rhythm	Sinus bradycardia, long PR (0.30 s)
POC glucose	104

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness (RASS) and orientation	Very sleepy, rouses only to a loud voice, falls back asleep mid-sentence (RASS -2); oriented to name only; mumbles "I... can't... keep my eyes open. Legs feel like lead."
Breathing: rate for a FULL minute, depth, pattern	RR 10, shallow and slow, no distress
Reflexes: patellar and biceps (0 to 4+)	ABSENT (0) at patellar and biceps
Strength: grip and lifting the legs	Can barely lift the legs off the bed; grip very weak
Lungs and neck veins	Lungs clear
Skin, mouth and pupils	Flushed and warm; pupils 3 mm and reactive (not an opioid picture: none since Monday)
Urine (Foley): color and amount	Dark amber; last hour 40 mL (0700-1100 hourly 35, 30, 55, 40 = 160 mL); the guide has the instructor say "about 20 mL/h": accept 20-40 mL/h as low

Skip order if behind: 2B3, 2A3. Safety/SBAR point: yes (1).

2C1 Call for help?

Look at your numbers and your exam. Is this a "call for help NOW" moment? Name the numbers or findings that tell you.

Model answer: Yes: RR 10 shallow, HR 44, SBP 84, SpO2 91% on room air, reflexes now absent, less alert (rouses to a loud voice only). That is every call parameter in the vitals order (RR under 12, HR under 45, SBP under 90, reflexes lost, drop in alertness). Call the rapid response team and stay with the patient.

Why (tied to the chart): Loss of reflexes came first (0700 1+, now 0); breathing is now failing. Do not wait to finish other tasks.

Look for: Full credit = a decision (call RRT) plus at least three findings.

Common misses: Waiting for the next scheduled check; thinking "SpO2 91% is only a little low".

2C2 The oxygen trap

You put Ray on oxygen and the SpO2 climbs. Did that fix the problem? Why or why not?

Model answer: No. Oxygen raises the SpO2 number but does not make weak breathing muscles breathe faster or deeper. His RR is still 10 and shallow; CO2 builds up. Watch RR, depth and drowsiness (and capnography), and be ready with suction and a bag-valve-mask.

Why (tied to the chart): The 1130 ABG (7.24 / PaCO2 46) and 1415 ABG (7.20 / 52) show the CO2 rising even with oxygen on.

Look for: Look for "slow shallow breathing is the problem, not low oxygen".

Common misses: Feeling reassured by a good SpO2.

2A1 Call numbers

Find the nursing order for vital signs, level of consciousness, RR and reflexes. Copy its "call the provider for..." list. How many of those are true for Ray right now?

Model answer: Call for RR under 12, HR under 45, SBP under 90, reflexes lost, or any drop in alertness. All five are true right now (RR 10, HR 44, SBP 84, reflexes 0, rouses only to loud voice).

Why (tied to the chart): Order from Dr. Lindgren, 0639. The orders already told the nurse when to call.

Look for: Full credit = list copied and an honest count.

Common misses: Copying only one parameter.

2A2 The antidote order

Find the PRN calcium gluconate order. Write: dose, how fast, and what the nurse must check before and during.

Model answer: 1 g IV over 10 minutes, on the cardiac monitor, through a patent large vein (the 18 g right antecubital). Check the site and blood return before and during (calcium injures tissue if it leaks); watch HR, BP and rhythm; do not mix with other drugs in the line.

Why (tied to the chart): Calcium directly opposes magnesium at the nerve-muscle junction and heart for 30 to 60 minutes. It does not lower the magnesium.

Look for: Accept dose, 10 minutes, monitor, vein/site.

Common misses: Pushing it fast; no monitor; forgetting the IV site check.

2A3 What's missing?

A deteriorating patient needs more than Ray has in the chart right now. List 3 things that are NOT yet ordered or set up and could save his life.

Model answer: Any three: oxygen order and continuous capnography (EtCO2); pacing pads on and connected; atropine as a back-up; vitals every 15 minutes; rapid response called; STAT repeat magnesium, ABG and 12-lead ECG; urgent nephrology and ICU outreach; stop the maintenance saline until reviewed.

Why (tied to the chart): These are exactly what the rapid response orders add at 1115 (calcium 1 g, NS 250 mL, oxygen 2-4 L, atropine PRN, pacing pads, vitals q15, STAT labs).

Look for: Count any three reasonable safety items.

Common misses: Listing meds only; forgetting equipment (pads, suction, BVM).

2B1 Update the labs

Copy the ED 0440 values (from your last page) and the new 0900 values. Draw an arrow: up, down or same.

Test	Tue 0440	Tue 0900	Arrow
Magnesium	7.8	8.7	up
Potassium	5.4	5.3	about same
Creatinine	4.9	4.8	about same
Calcium	8.1	8.7	up (normal now)

Model answer: Only magnesium moved the wrong way.

Why (tied to the chart): Creatinine did not improve even after fluids: the kidneys are not "waking up" yet.

Look for: Score on mostly correct values and arrows.

Common misses: Copying 0900 values from the wrong row.

2B2 Why is it still rising?

Ray got about a liter of saline and a big dose of furosemide, yet Mg went up (7.8 to 8.7). Also read the new urine test: FENa 0.6% (under 1% = "prerenal"). What does all this say about why the magnesium is not leaving?

Model answer: His kidneys are not making enough urine (hourly output only 30-55 mL, then lower) and are barely filtering (creatinine unchanged, eGFR 12). FENa 0.6% means the kidneys are holding on to sodium because he is dry, so saline and furosemide only help if the kidneys can respond. The ultrasound shows no blockage. Magnesium keeps coming from the gut, too.

Why (tied to the chart): Chart: UO 35, 30, 55, 40 mL/h; creatinine 4.9 to 4.8; FENa 0.6%; renal ultrasound (1015): no obstruction.

Look for: Full credit = kidneys cannot clear it + at least one lab clue.

Common misses: Saying "needs more fluid" (heart failure and low urine make that risky).

2B3 Rhythm check (stretch)

The monitor now shows PR 0.30 s (0700: 0.26; ED ECG: 248 ms). Longer or shorter? Which strip on the Handoff page is it heading toward? What do you want the team to have ready?

Model answer: Longer: the conduction delay is growing, heading toward strip C (wide QRS) and eventually heart block. Have a 12-lead ECG, pacing pads and calcium gluconate ready; atropine is the back-up.

Why (tied to the chart): ECG at 1130 shows PR 300, QRS 136 (appears after the team acts).

Look for: Accept any ready-equipment answer.

Common misses: Saying the rhythm is "fine, just slow".

2M Mg-o-meter

New level (Tue 0900): _____ mg/dL Zone: _____

Model answer: X at 8.7 (Tue 0900), in the REFLEXES LOST band (7-10), matching the absent reflexes at the bedside.

Why (tied to the chart): Up from 7.8: the trend, not just the number, is the alarm.

Look for: Means: reflexes lost, very sleepy, slow HR and ECG changes, close to the breathing-depression zone.

Common misses: X in the wrong column.

2T1 Top 3 priorities

Top 3 priorities right now, in order. (Think airway, breathing, circulation.)

Model answer: 1) Call the rapid response team NOW (RR 10, HR 44, BP 84/48, SpO2 91%, reflexes gone, less alert) and stay with Ray. 2) Airway and breathing: HOB up, stimulate, oxygen but keep watching RR/CO2, suction and bag-valve-mask at the bed, strict NPO. 3) Heart and antidote: monitor and 12-lead, pacing pads, calcium gluconate 1 g IV over 10 minutes on the monitor. Also: no sedatives or opioids, hold maintenance saline until reviewed, update Eleanor.

Why: ABC first; calcium is first-line, atropine and pacing are back-ups.

Look for: Full credit = calling for help is in the top 3 and airway/breathing is addressed.

Common misses: Starting with labs or paperwork; not calling until more data come in.

2T2 One question for the provider

The ONE question for the rapid response doctor or hospitalist:

Model answer: Examples: "Do you want calcium gluconate 1 g now?" "Can nephrology and ICU outreach see him urgently, and what is the plan if his breathing gets slower?" "Should I stop the maintenance saline?"

Look for: Specific, clinical and actionable.

Common misses: Asking something already answered by the orders.

2T3 SBAR

SBAR call to the rapid response team / hospitalist.

Model answer: S: "RN for Raymond Kessler, 5E-14. I am calling a rapid response: RR 10 and shallow, HR 44, BP 84/48, SpO2 91% on room air, only waking to a loud voice, reflexes gone." B: "74-year-old with CKD 4 admitted with Mg 7.8, calcium gluconate in the ED; 0900 Mg was 8.7. Home magnesium products were found. Urine low, heart failure, NPO." A: "I think magnesium is still rising and he has respiratory depression and heart block risk. He is not protecting his breathing well." R: "I need you at the bedside now. I would like calcium gluconate 1 g IV, oxygen, STAT Mg and ECG, nephrology and ICU on the phone, pacing pads at the bedside."

Why: Matches the model SBAR #1 in INSTRUCTOR_GUIDE.md section 6.

Look for: Look for "I am calling a rapid response", numbers and a clear request.

Common misses: Calling without numbers; asking only "Can you come look at him?"

2P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #2**: Calcium protects, it does not remove. It buys 30 to 60 minutes while magnesium is held back from the nerves and heart. Something has to take the magnesium OUT of the body. What could that be?

CHECKPOINT 3: THE BIG DECISION

Tuesday 1500 | 4 student pages

Read-aloud Situation (also printed in the packet):

It is 1500. The rapid response team came at 1115. Ray got calcium gluconate (1 g at 1120, and 2 g more ordered at 1235 and again at 1400). Nephrology ordered a big dose of furosemide, but he made only 70 mL of urine in four hours.

At 1400 the lab called: magnesium 10.9, potassium 5.8, creatinine 5.3. The 1330 ECG showed HR 38 and a wider QRS. Dr. Raman decided on emergency dialysis. Eleanor gave phone consent at 1350 with two nurses listening. The dialysis catheter went into the right side of his neck at 1440 and the x-ray is clear. Dialysis starts now. ICU outreach says he can stay on 5 East with a 1:1 nurse.

Set the monitor to	Value
Temp (F)	97.2
HR	46
BP (MAP)	88/50 (63)
RR	9
SpO2 / O2	92% on 4 L nasal cannula
EtCO2	52
Pain	CPOT 0
Rhythm	Sinus bradycardia, long PR (0.32 s), wide QRS (0.14 s), tall T waves
POC glucose	88

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness (RASS) and orientation	Responds to a loud voice only, opens eyes briefly (RASS -3); cannot follow commands reliably; airway intact, gag present
Breathing: rate, depth, EtCO2	RR 9, shallow; EtCO2 52 mmHg; SpO2 92% on 4 L
Reflexes: patellar and biceps (0 to 4+)	ABSENT (0)
Strength: grip and lifting the legs	Flaccid weakness: cannot lift a hand off the bed
Lungs and neck veins	NEW fine crackles at both bases; neck veins flat to mildly up; trace ankle edema
Neck catheter site	Right IJ dialysis catheter dressing dry, no bleeding
Urine (Foley): color and amount	Dark amber; last hour 10 mL (1100-1500 hourly 20, 25, 15, 10 = 70 mL)

Skip order if behind: 3B4 + diagram, 3A3. Safety/SBAR point: yes (1).

3C1 Breathing watch

Write your RR, SpO2 and EtCO2 trend (0700, 1100, now). If Ray's RR drops to 7 during dialysis, what do you do in the first minute?

Model answer: Trend: RR 14, then 10, now 9; SpO2 94, 91, now 92 on 4 L; EtCO2 52. If RR 7: stimulate, call the rapid response/ICU outreach immediately (RR under 8 is a transfer/airway criterion), bag-valve-mask and suction ready, airway cart in the room, tell the dialysis RN and provider.

Why (tied to the chart): ICU outreach note (1435): intubate and transfer if RR under 8, PaCO2 over 60 or pH under 7.15, SpO2 under 90% on 6 L, decline in alertness or loss of airway reflexes.

Look for: Full credit = trend noted and a first-minute plan.

Common misses: Waiting to see if it comes back up.

3C2 New crackles

Crackles at both bases are NEW since 0700. What do you suspect? What do you not want to give, and who do you tell?

Model answer: Fluid overload (heart failure plus saline: about 1.4 L net positive since 0700; the 1455 chest x-ray shows mild interstitial prominence). Do not give more IV fluid or run the maintenance saline (it was stopped at 1415). Tell the provider and the dialysis nurse; dialysis can pull fluid off.

Why (tied to the chart): I&O: in 800 + 845, out 160 + 70. Sets up why the heart failure matters when fluids are used as treatment.

Look for: Full credit = overload + no fluids + who to tell.

Common misses: Treating crackles with more fluid; ignoring them because of the bigger crisis.

3A1 Dialysis readiness check

For each item write where you found it in the chart, or mark it missing.

Check before the machine starts	Found? Where?
Consent: Eleanor (health care proxy) by phone at 1350, two RN witnesses	Nephrology HD decision note 1355; Procedure note 1450
HD 3 hours, low-magnesium dialysate, heparin-free, ultrafiltration as BP tolerates	Orders 1350
Right IJ catheter placed 1440; chest x-ray 1455 clear	Procedure note 1450; CXR 1455; Devices
Calcium gluconate 2 g IV over 20 minutes, due 1400	Orders/MAR 1350
NS 100 mL/h stopped 1415	MAR/Orders
Held: lisinopril and metoprolol (home meds held on admission); no antihypertensives during dialysis	Orders (held home meds); HD order note

Model answer: All six exist in the chart.

Why (tied to the chart): A pre-procedure checklist is what keeps a rushed procedure safe.

Look for: Credit each item that is correctly located; "missing" is fine only if it truly is not there (none of these is missing).

Common misses: Not realizing the chest x-ray must be read before the catheter is used.

3A2 Catheter rules

The new neck catheter is for dialysis. Write two things you may do and two things you must never do with it.

Model answer: May: check the dressing is dry and intact; watch for bleeding, redness, drainage; monitor BP and rhythm during dialysis; keep it covered and secure. Never: flush it, draw labs from it, give IV medicines or fluids through it (dialysis staff only; heparin lock per protocol).

Why (tied to the chart): Order 1455: use by dialysis staff only; never flush or draw labs.

Look for: Two and two.

Common misses: Using it "just this once" for the calcium.

3A3 ICU thresholds

Find the ICU outreach note (written 1435). Copy three numbers or findings that mean "intubate and transfer to the ICU".

Model answer: Any three of: RR under 8; PaCO2 over 60 or pH under 7.15; SpO2 under 90% on 6 L; decline in alertness; loss of airway reflexes. (Also: no BiPAP; keep bag-valve-mask, suction and airway cart in the room.)

Why (tied to the chart): ICU outreach note, H. Moreno MD, 1435.

Look for: Credit three correct items.

Common misses: Writing numbers from the ABG instead of the thresholds.

3B1 Trend the labs

Copy the three Mg draws and the other numbers. Then circle the one that worries you most.

Test	Tue 0900	Tue 1115	Tue 1315
Magnesium	8.7	9.5	10.9
Potassium	5.3	5.6	5.8
Creatinine	4.8	5.0	5.3
CO2	18	17	16

Model answer: All four are going the wrong way. The Mg of 10.9 (resulted 1400) is the headline.

Why (tied to the chart): Everything rises together because the kidneys are failing and nothing is removing it.

Look for: Mostly correct values; a sensible circled item (Mg 10.9, K 5.8 for the heart, or CO2 16).

Common misses: Using collected vs result times inconsistently (1115 draw resulted at 1200; 1315 draw at 1400).

3B2 Blood gas and breathing

Copy the two blood gases. What do they tell you about Ray's breathing, and why is oxygen not the fix?

ABG	Normal	1130 (on 3 L)	1415 (on 4 L)
pH	7.35-7.45	7.24	7.20
PaCO2	35-45	46	52
PaO2	80-100	72	68

HCO3	22-26	19	20

Model answer: PaCO2 rose and pH fell: he is not breathing out enough CO2 (hypoventilation, respiratory acidosis) even with oxygen on. Oxygen does not strengthen weak breathing muscles; the magnesium level has to come down.

Why (tied to the chart): Sets up why dialysis is urgent; also why RR and EtCO2 beat SpO2.

Look for: Credit for values and the CO2 explanation.

Common misses: Reading the PaO2 only.

3B3 Heart strip

ECG at 1330: HR 38, PR 320 ms, QRS 148 ms, tall peaked T waves. Which strip (A, B or C)? Name two reasons T waves can be tall in Ray (hint: two labs).

Model answer: C. Tall peaked T waves can come from high magnesium (conduction trouble) and from high potassium (5.8 at 1315). Both are high in Ray.

Why (tied to the chart): ECG 1330 note: "tall peaked T waves (K 5.8)".

Look for: C and magnesium + potassium.

Common misses: Forgetting potassium; it needs dialysis too.

3B4 Label the machine (stretch)

Name parts 1 to 4 from the word bank, then finish: "Magnesium leaves the blood because...". Word bank: blood pump / dialyzer filter (membrane) / cleaned blood returns / fresh bath low in magnesium

#	What it is
1	blood pump
2	dialyzer filter (membrane)
3	fresh bath (dialysate) LOW in magnesium
4	cleaned blood returns to Ray
Why?	Magnesium moves from the blood (high) across the membrane into the bath (low) until they are close; the used bath is drained away.

Model answer: See table.

Why (tied to the chart): Dialysis removes magnesium directly; nothing else can when the kidneys are not working (the 3-hour session takes Mg 10.9 to 4.6).

Look for: All four labels and a "high to low" idea.

Common misses: Thinking the machine adds something rather than removing.

3M Mg-o-meter

New levels (1115 and 1315): _____ and _____ mg/dL Zone for the higher one: _____

Model answer: X at 9.5 (1115) in REFLEXES LOST; X at 10.9 (1315) in the top DANGER band (over 10): breathing depression and heart block. Connect the dots.

Why (tied to the chart): Matches RR 9, EtCO2 52, QRS 148. Peak of the stay.

Look for: Means: the numbers have crossed into the breathing-depression zone.

Common misses: Plotting the result time instead of the draw time is fine as long as order is right.

3T1 Top 3 priorities

Top 3 priorities while dialysis starts, in order.

Model answer: 1) Airway and breathing: RR, EtCO2, SpO2, HOB 30-45, suction, bag-valve-mask and airway cart in the room; know the ICU outreach triggers (RR under 8, PaCO2 over 60, SpO2 under 90% on 6 L, decline in alertness). 2) Dialysis safety: consent confirmed, time-out/ID, catheter for dialysis staff only, BP and HR every 15 minutes, keep calcium on the cart, hold BP medicines. 3) Heart and fluid: calcium gluconate 2 g per order on the monitor, watch crackles and strict I&O (urine 10 mL/h).

Why: Order logic: the airway is the thing that can still kill him in the next hour.

Look for: Accept a different order if #1 is defended; airway/breathing and dialysis safety should both appear.

Common misses: Putting family updates first; forgetting the airway because "dialysis will fix it".

3T2 One question for the provider

The ONE question for the ICU outreach team / nephrology:

Model answer: Examples: "What exact numbers make you want him in the ICU, and who do I call?" "Do you want a repeat ABG during dialysis?" "What do you want me to do if his BP drops during dialysis?"

Look for: Specific and safety-focused.

Common misses: A question about discharge or diet.

3T3 SBAR

SBAR call to ICU outreach / nephrology (his breathing is slowing before dialysis even starts).

Model answer: S: "RR 9, SpO2 92% on 4 L, EtCO2 52, HR 46, BP 88/50, only waking to a loud voice, dialysis about to start." B: "Mg 10.9 (1315), K 5.8, QRS 148 ms, ABG 7.20/52 at 1415, calcium gluconate 2 g due at 1400, dialysis catheter placed 1440, chest x-ray clear." A: "He is hypoventilating and may need airway support before dialysis finishes." R: "Please come now or tell me your thresholds. I would like the airway cart in the room, a repeat ABG and a decision about transfer."

Why: Matches model SBAR #3.

Look for: Numbers, a clear assessment and a request with a timeframe.

Common misses: No recommendation/ask.

3P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #3**: Dialysis is the exit door that works when the kidneys cannot. It can drop magnesium fast, but fast changes can cause their own problems. Keep your detective eyes open for what a 3-hour session might do to the body.

CHECKPOINT 4: THE QUIET AFTER THE STORM

Tuesday 1900 | 3 student pages

Read-aloud Situation (also printed in the packet):

It is 1900. Dialysis ran from 1500 to 1800 and took off 500 mL of fluid. At 1630 his blood pressure dipped to 82/46; the dialysis nurse gave 250 mL of saline and slowed the machine. By 1700 his magnesium was 7.2 and after dialysis (1800) it was 4.6.

Ray is awake and breathing 16 times a minute. The 1850 results are in. Your squad rolls straight into the night shift as the day team gives report. Look him over.

Set the monitor to	Value
Temp (F)	97.6
HR	66
BP (MAP)	104/60 (75)
RR	16
SpO2 / O2	96% room air
EtCO2	not shown
Pain	3/10 back and neck catheter site
Rhythm	Normal sinus rhythm 66, PR 0.20 s
POC glucose	54 (at 1905)

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness and orientation	Awake, oriented to person, place and time, a little tired; speech clear
Breathing: rate for a FULL minute, depth, pattern	RR 16, regular, normal depth
Reflexes: patellar and biceps (0 to 4+)	Returning: patellar 1+, biceps 1+; grip stronger
Lungs and swallow screen	Not scripted: instructor decides. Fine crackles should be fewer after 500 mL ultrafiltration. Passes a bedside swallow screen (sips water, no cough)
Mouth, hands and how he feels	At 1905: "I feel shaky... I'm sweating... I feel funny around my mouth and my hands tingle." Skin cool and damp, tremor
Neck catheter site	Dialysis catheter dressing dry and intact
Urine (Foley): color and amount	last hour 20 mL (1500-1900 hourly 15, 20, 20, 20 = 75 mL; plus 500 mL pulled off by dialysis)

Skip order if behind: 4B3, 4A3. Safety/SBAR point: yes (1).

4C1 Post-dialysis watch

The dialysis RN says: "Check Ray every 15 minutes for the next hour." Name four things you will check or protect.

Model answer: BP and HR (dialysis can drop BP), rhythm, RR, reflexes and LOC, glucose, catheter dressing (bleeding), and tingling/cramps (calcium). Orders: vitals and glucose every 15 minutes x4, then hourly x4.

Why (tied to the chart): Post-dialysis order 1845.

Look for: Any four with at least BP/HR and glucose.

Common misses: Only checking vital signs.

4C2 If glucose is low

When you check his glucose at 1905 the instructor tells you the number. Write the number, the steps you take, and what you chart.

Model answer: POC glucose 54 (shaky, sweating, awake and swallowing safely). Rule of 15: confirm with the meter, swallow screen, 15 g carbohydrate by mouth (4 oz apple juice or glucose tablets, NOT orange juice because of his potassium), recheck in 15 minutes (68), repeat 15 g (recheck 84), then a snack; call the provider; if he could not swallow or was not alert: D50 25 g IV through the 18 g right antecubital. Chart times and values.

Why (tied to the chart): Insulin lasts longer in kidney failure (glargine 22 units Monday), almost no food for 4 days, dialysis. Serum glucose was 71 at 1700 (the warning).

Look for: Look for swallow screen BEFORE juice, juice choice, 15-minute recheck, call.

Common misses: Orange juice; waiting 30+ minutes to recheck; no swallow screen.

4A1 The glucose trend

Copy the serum glucose at each draw. Then answer: was any insulin given? What did the numbers predict?

Tue 0440	Tue 0900	Tue 1115	Tue 1315	Tue 1700
148	122	108	96	71

Model answer: 148, 122, 108, 96, 71: a steady fall with NO insulin given (glargine held, correction scale not triggered).

Why (tied to the chart): Hidden trend: a glucose of 71 at 1700 was the warning.

Look for: All five values.

Common misses: Missing the 1700 value (lab draw during dialysis).

4A2 Insulin orders

Which insulin orders are active right now? What do you want the provider to hold or change, and why?

Model answer: Lispro renal-sensitive correction scale every 4 hours is active (glargine is held; last home dose 22 units Monday 2100). Ask to hold correction insulin (for example if glucose under 100) and for dextrose and hourly glucose checks, and a diet order. Reason: insulin lasts longer in kidney failure and he has eaten almost nothing for 4 days.

Why (tied to the chart): The hold-under-100 order appears at 1914 after the event.

Look for: Full credit = names the scale + a hold request + reason.

Common misses: Asking to restart glargine.

4A3 Protocol and juice

Find the nurse-driven hypoglycemia protocol order. Write its steps. Then name one juice that is a poor choice for Ray, and why.

Model answer: Rule of 15: 15 g carbohydrate by mouth, recheck in 15 minutes, repeat if under 70, then a snack once over 70 (awake and able to swallow only); D50 25 g IV if not. Poor choice: orange juice (high in potassium; his K was 5.4-5.8 earlier). Apple, cranberry or grape juice are lower in potassium.

Why (tied to the chart): Orders: hypoglycemia protocol and dextrose 50%.

Look for: Steps and one correct juice reason.

Common misses: Writing the dose of insulin to give.

4B1 After the machine

Copy the numbers. Which ones improved? Which one is now too low?

Test	Tue 1315	Tue 1700	Tue 1800
Magnesium	10.9	7.2	4.6
Potassium	5.8	4.4	3.9
Creatinine	5.3	3.9	3.0
Calcium	8.3	8.0	7.7
Ionized calcium	(not drawn)	(not drawn)	1.02

Model answer: Mg, K and creatinine improved. Calcium (7.7) and ionized calcium (1.02) are now too low.

Why (tied to the chart): Low calcium can cause tingling around the mouth, cramps and a longer QTc.

Look for: Values + calcium identified as the low one.

Common misses: Calling a post-dialysis Mg of 4.6 "normal" (it is still above 2.2).

4B2 Calcium clue

Ray says his mouth and hands tingle. The 1830 ECG shows QTc 486. How do those fit together? What would you report, and when would calcium be given?

Model answer: Low calcium (7.7, ionized 1.02) causes tingling, cramps and a long QTc; the dialysis bath removed calcium and magnesium, and low magnesium can also lower parathyroid hormone. Report the symptoms and the QTc; calcium only if symptomatic or QTc over 500 per nephrology.

Why (tied to the chart): ECG 1830: sinus 66, PR 204, QRS 108, QTc 486. Nephrology note 1850.

Look for: Link between low calcium and symptoms/QTc.

Common misses: Giving calcium right away without an order.

4B3 Rhythm check (stretch)

Look at the monitor rhythm (PR 0.20 s) and the 1830 ECG (PR 204, QRS 108). Which strip is it most like now, and what got better?

Model answer: Strip A (normal): PR near 200 ms and QRS 108 ms (normal) are back; the conduction delay and wide QRS cleared as magnesium fell.

Look for: A with improvement noted.

Common misses: Choosing B because PR is "204".

4M Mg-o-meter

New levels (1700 and 1800): _____ and _____ mg/dL Zone for the lower one: _____

Model answer: X at 7.2 (1700) in REFLEXES LOST (just inside); X at 4.6 (1800) in the EARLY band (2.3-5). Big drop in three hours.

Why (tied to the chart): Dialysis removed magnesium fast. 4.6 is still above normal.

Look for: Means: out of the danger zone but not normal.

Common misses: Writing "normal".

4T1 Top 3 priorities

Top 3 priorities right now, in order.

Model answer: 1) Safe treatment of the low glucose (swallow screen, 15 g carbohydrate, recheck in 15 min) and call the provider. 2) Post-dialysis checks every 15 minutes: BP, HR, rhythm, RR, reflexes, LOC, catheter dressing, tingling. 3) Hand off to the night team: glucose event, calcium and QTc watch, dialysis catheter, expect magnesium to creep back up, bowel plan.

Why: Hypoglycemia is the immediate threat (seizure, harm); calcium and rebound are the follow-ups.

Look for: Different order accepted if glucose and post-dialysis monitoring are both present.

Common misses: Forgetting to recheck glucose; skipping the handoff.

4T2 One question for the provider

The ONE question for the provider:

Model answer: Examples: "Can I get a D10W order, hourly glucose checks and a plan to hold correction insulin?" "Which insulin do you want continued, if any?"

Look for: Specific to the glucose problem.

Common misses: Asking for more insulin.

4T3 SBAR

SBAR call to the hospitalist about the low glucose.

Model answer: S: "Mr. Kessler's glucose is 54, he is shaky and sweating, awake and oriented." B: "74-year-old with CKD 4, dialysis finished at 1800. No insulin since Monday night (glargine 22 units); serum glucose was 71 at 1700; almost no food for 4 days. I gave 15 g apple juice at 1905 after a swallow screen; at recheck it was 68, so I gave 15 g more; then 84." A: "Insulin is lasting longer from the kidney failure, plus dialysis and no intake." R: "Can I have a D10W order, hourly glucose, a plan to hold correction insulin and a diet order? Please review all his insulin."

Why: Matches model SBAR #4.

Look for: Treatment given and results included.

Common misses: No numbers for the rechecks.

4P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #4**: Magnesium is not only in the blood. A lot of it hides in cells, bone and the bowel. When the blood level drops fast, stored magnesium can drift back out. Remember this: the number can come back.

CHECKPOINT 5: THE NIGHT SHIFT RIDDLE

Tuesday night 2300 | 3 student pages

Read-aloud Situation (also printed in the packet):

It is 2300. D10W (sugar water) has run since 1930 at 40 mL/h and Ray's sugars have been 98, 106 and 118. At 2100 his magnesium was 5.6, up from 4.6, and his calcium was 7.9. Eleanor went home at 2100.

Since 2230 Ray has been restless and is pulling at his lines. You go in.

Set the monitor to	Value
Temp (F)	98.2
HR	62
BP (MAP)	112/64 (80)
RR	16
SpO2 / O2	96% room air
EtCO2	not shown
Pain	3/10 back
Rhythm	Sinus rhythm 62, PR 0.22 s
POC glucose	118

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness; CAM features (see box)	Awake, restless, confused, thinks it is a work day at the machine shop; briefly redirectable. CAM POSITIVE. "I've got to get to the shop. Get these things off me! Where is Eleanor?"
Breathing: rate for a FULL minute, depth, pattern	RR 16, no distress
Reflexes: patellar and biceps (0 to 4+)	1+ (still diminished)
Strength / moving in bed	Tries to get out of bed; moves all limbs (not scripted beyond this)
Lungs and neck veins	Fine crackles at the bases are clearing
Lines and tubes (what is he pulling at?)	Picking at the Foley and the neck dressing
Urine (Foley): color and amount	last hour 60 mL (1900-2300 hourly 45, 50, 55, 60 = 210 mL); the guide reads "55 mL/h": accept 45-60

Skip order if behind: 5B3, 5A3. Safety/SBAR point: yes (1).

5C1 Safe and calm

List five non-drug things you will do right now to keep Ray safe and calm, including how you protect his two lines (Foley and neck catheter).

Model answer: Any five: reorient gently; bed low, bed alarm, call light; 1:1 sitter or frequent rounding; lights low and quiet; call Eleanor (ask her to come back or talk on the phone); cover/secure the Foley and the neck dressing; toileting and pain check; glasses or hearing aids if used; cluster care; avoid restraints.

Why (tied to the chart): Delirium precautions order (2304): no restraints, no sedatives.

Look for: Five items with a line-protection idea.

Common misses: Restraints or mittens as the first answer; "give something to calm him".

5C2 Do not hide the clues

Even with delirium, which three checks do you keep doing (they warn of magnesium trouble)? Why can sleepiness hide them?

Model answer: RR, reflexes and heart rhythm/HR (and level of alertness). A sedated or sleepy patient may look "peaceful" while breathing is slowing and reflexes are fading. Confusion can also hide early respiratory depression.

Why (tied to the chart): The 2300 Mg is rising (5.6); post-dialysis rebound is possible.

Look for: RR + reflexes + rhythm.

Common misses: Stopping reflex checks because he is agitated.

5A1 Suspect medicines

Scan the Orders and MAR for anything that could cause or worsen confusion, or hide breathing and reflex checks. Name each one and say what you would ask the provider to do.

Model answer: Diphenhydramine 25 mg PO at bedtime PRN for sleep (admission order set): anticholinergic and sedating, on the Beers list, bad in age 74, kidney failure and magnesium toxicity. Do not give; ask to discontinue. Also keep gabapentin and oxycodone held; avoid benzodiazepines and zolpidem. Melatonin is the safer ask.

Why (tied to the chart): Chart: diphenhydramine order from the default admission order set (ordered 0639).

Look for: Names diphenhydramine and asks to stop it.

Common misses: Missing it because it is a PRN buried in the MAR.

5A2 Reversible causes checklist

For each cause, write what the chart or monitor says in a few words.

Cause of delirium to check	What the chart says
Glucose	POC 118 now (98, 106, 118 on D10W): not low
Bladder / Foley; pain	Draining: 45, 50, 55, 60 mL/h; check for kink. Pain 3/10 back
Bowel	No stool since Fri 11/13 (assessment 0700); PEG and senna due 2200
Oxygen	SpO2 96% on room air
New medicines	Diphenhydramine PRN in the order set (not to be given); D10W; melatonin not yet ordered
Labs	Mg 5.6 (rising from 4.6); calcium 7.9, ionized 1.06 (low); creatinine 3.4 (uremia can confuse)

Model answer: Not low glucose. Multifactorial: hospital night, uremia, recent hypoglycemia, magnesium/calcium shifts, constipation.

Why (tied to the chart): A checklist stops you from reaching for a sedative.

Look for: Credit when most rows cite a real chart fact.

Common misses: Writing "normal" without a number.

5A3 Bowel plan

PEG and senna were ordered at 2130. Why does constipation matter in a magnesium case, and which two kinds of products must stay away?

Model answer: Unabsorbed magnesium sitting in the bowel keeps being absorbed, and Ray has had no stool since Friday. Use non-magnesium laxatives (polyethylene glycol, senna, bisacodyl). Stay away from magnesium laxatives/antacids (milk of magnesia, magnesium citrate, Epsom salt) and phosphate (Fleet) enemas.

Why (tied to the chart): Orders: PEG 17 g BID and senna 17.2 mg HS (2130); NO magnesium/phosphate precaution.

Look for: Gut-absorption reason + two product families.

Common misses: Only "to make him comfortable".

5B1 New labs

Copy the 1800 (after dialysis) and 2100 values. Draw an arrow for the change.

Test	Tue 1800	Tue 2100	Arrow
Magnesium	4.6	5.6	up
Potassium	3.9	4.3	up (normal)
Creatinine	3.0	3.4	up
Calcium	7.7	7.9	about same (low; ionized 1.02 to 1.06)

Model answer: Magnesium and creatinine have started to climb again; calcium remains low.

Why (tied to the chart): Creatinine rising after dialysis is expected as the machine effect fades.

Look for: Correct values and arrows.

Common misses: Using 1700 instead of 1800.

5B2 What is going on with Mg?

Magnesium went 10.9, then 4.6 (1800), now 5.6. Explain what you think is happening in plain words. Where could the extra magnesium be coming from?

Model answer: Rebound: after dialysis, magnesium shifts back into the blood from cells, bone and the bowel, and the kidneys (creatinine 3.4) still cannot clear it. Leftover magnesium in the gut (constipation, no stool since Friday) is also being absorbed.

Why (tied to the chart): Mg 4.6 to 5.6 to (6.4 at 0100). Night hospitalist note names this a rebound.

Look for: Full credit = rebound idea + at least one source (cells/bone/gut) + weak kidneys.

Common misses: Saying the lab made an error or the dialysis did not work.

5B3 Which numbers could be muddling his brain? (stretch)

Pick two labs from your table that could add to confusion, and say how.

Model answer: Creatinine/BUN (uremia: 3.4 and 40 still high), calcium (7.9, ionized 1.06 is low), magnesium (5.6 and rising). Any two with a reason.

Look for: Two labs with reasons.

Common misses: Picking potassium (4.3 is normal).

5M Mg-o-meter

New level (Tue 2100): _____ mg/dL Zone: _____

Model answer: X at 5.6 (2100) in the FLUSHING band (5-7).

Why (tied to the chart): Up one point after dialysis: the line is bending upward again.

Look for: Means: drowsy, low BP, weak reflexes possible again; keep checking RR and reflexes.

Common misses: Plotting 4.6.

5T1 Top 3 priorities

Top 3 priorities right now, in order.

Model answer: 1) Safety: protect Ray and his lines (bed alarm, sitter or frequent rounding, secure Foley and neck dressing, low stimulation, no restraints). 2) Look for reversible causes and keep NOT giving sedatives (no diphenhydramine, benzodiazepines, zolpidem or opioids); keep RR, reflex and rhythm checks going. 3) Call the night hospitalist (SBAR); ask to discontinue the sleep-aid order, add melatonin and delirium precautions, and recheck magnesium; bowel regimen.

Why: The sedative trap is the lesson: it can mask the breathing and reflex checks in a rebounding magnesium.

Look for: Credit if safety, no-sedative and a call are present.

Common misses: Giving the PRN sleep medicine; restraints.

5T2 One question for the provider

The ONE question for the night hospitalist:

Model answer: Examples: "Can we discontinue the diphenhydramine sleep order and use melatonin instead?" "Can we check the magnesium sooner than 0100?" "Can Eleanor come back or can we have a sitter?"

Look for: Concrete order request.

Common misses: Asking "Can I give something to help him sleep?"

5T3 SBAR

SBAR call to the night hospitalist about new confusion.

Model answer: S: "Mr. Kessler is newly confused and restless, pulling at his lines, CAM positive. Vitals are stable and glucose is 118." B: "Dialysis today for Mg 10.9; Mg 5.6 at 2100 and rising; calcium 7.9. No stool for 4 days. The order set has diphenhydramine for sleep, which I am not giving." A: "I think the delirium is from hospital, uremia and possibly rebound magnesium. I worry sedation would mask his breathing and reflexes." R: "Can we discontinue diphenhydramine, avoid benzodiazepines, order melatonin and delirium precautions, a sitter, and recheck the magnesium sooner?"

Why: Matches model SBAR #5.

Look for: Names what was NOT given and why.

Common misses: No assessment (the "A").

5P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #5**: The riddle has a hidden accomplice: a "helpful" order that looks routine. Routine orders and standing protocols are written for ordinary patients. Always read the ingredients of every order in a kidney patient.

CHECKPOINT 6: THE BOOMERANG

Wednesday 0300 | 3 student pages

Read-aloud Situation (also printed in the packet):

It is 0300 Wednesday. Ray slept a little. The 0100 magnesium came back 6.4 (4.6 after dialysis, 5.6 at 2100). Calcium is 8.1. The D10W stopped at 0130 and his glucose is 126.

Ray has had no stool in four days and his belly looks full. You go to check him. He mumbles something about "the pink milk".

Set the monitor to	Value
Temp (F)	98.0
HR	56
BP (MAP)	104/60 (75)
RR	12
SpO2 / O2	95% room air
EtCO2	not shown
Pain	4/10 abdomen
Rhythm	Sinus rhythm 56, PR 0.24 s (long again)
POC glucose	126

Expected exam findings (what the Bedside Nurse should find and chart; portray these):

Check	Expected finding
Level of consciousness; CAM (see box)	Drowsy but wakes to voice, oriented to person and place, no longer agitated, CAM negative; speech slow
Breathing: rate for a FULL minute, depth, pattern	RR 12 (SpO2 drops to 93% only if asleep and not stimulated)
Reflexes: patellar and biceps (0 to 4+)	1+ (diminished again)
Strength: grip and lifting the arms	Weak but lifts his arms
Lungs and neck veins	Fine crackles at the bases (mild)
Abdomen: size, bowel sounds, tenderness	Distended, firm but not rigid, bowel sounds hypoactive, mild diffuse tenderness, no guarding
Urine (Foley): color and amount	last hour 40 mL (2300-0300 hourly 55, 50, 45, 40 = 190 mL)

Skip order if behind: 6A3, 6B3. Safety/SBAR point: yes (1).

6C1 "Can I have the pink milk?"

Ray says: "My belly is so full. Can I take something? At home I take the pink milk." What do you say and do? (Think: what is the pink milk, and what is in your protocol?)

Model answer: Do NOT give it. The pink milk is milk of magnesia (magnesium hydroxide). Explain kindly why (his kidneys cannot clear magnesium), check the abdomen, ask the provider for a safe bowel plan (bisacodyl suppository, polyethylene glycol already ordered, KUB), make sure the bowel protocol that ends in milk of magnesia is discontinued, and document.

Why (tied to the chart): Instructor-as-provider: "Good catch: do NOT give it. Discontinue the bowel protocol. Give bisacodyl 10 mg suppository. Get a KUB. No Fleet or magnesium enemas."

Look for: Says no, explains, and escalates for a safe alternative.

Common misses: Giving it because "it is the standing protocol"; a flat refusal with no alternative.

6C2 Stay ahead of the number

RR 12, HR 56, reflexes 1+. Compare with the call numbers in the vitals order (RR under 12, HR under 45, SBP under 90, reflexes lost, any drop in alertness). Is he there yet? How often will you recheck, and what makes you call a rapid response?

Model answer: Not yet: RR 12 is at the line, HR 56, SBP 104, reflexes 1+ and he wakes to voice. Recheck RR, reflexes, rhythm and LOC at least hourly; call a rapid response for RR under 10, HR under 45, SBP under 90, reflexes lost or a drop in alertness (from the 0320 plan).

Why (tied to the chart): Trend matters: RR was 16 at 2300 and is 12 now.

Look for: Names the trend and a recheck interval.

Common misses: Waiting for the parameter to be crossed before looking again.

6A1 Hidden magnesium hunt

Open the nurse-driven bowel protocol order and read it to the very end. Write each step, circle the step that is dangerous for Ray, and explain.

Model answer: Senna, then bisacodyl suppository, then milk of magnesia 30 mL if no stool in 72 hours. The last step is a magnesium product: dangerous in a patient whose kidneys cannot clear magnesium (Mg already 6.4 and rising).

Why (tied to the chart): Order ordered 0639, default set; discontinued 0330 after the RN questioned it. Sticky note and the active NO-magnesium precaution warn of exactly this.

Look for: Lists all three steps and names the last.

Common misses: Naming senna or bisacodyl as the problem.

6A2 Two orders disagree

Find the active precaution order about magnesium and phosphate. Both it and the bowel protocol are active. What do you do about this conflict, and who do you call?

Model answer: The precaution says NO magnesium-containing products (milk of magnesia, Maalox, magnesium oxide, citrate, Epsom salt) and no Fleet enemas. The protocol contradicts it. Do not give the protocol's last step; call the provider and ask to discontinue the protocol and replace it with a magnesium-free plan.

Why (tied to the chart): Order set clash: a safety order versus a default set.

Look for: Does not give + calls + names the conflict.

Common misses: Assuming the protocol must be right because it is a standing order.

6A3 A safe bowel plan (stretch)

List what is safe for Ray's bowels and what is not. Include one enema.

Model answer: Safe: polyethylene glycol (MiraLAX, ordered), senna, bisacodyl suppository (once ordered); mobilize as safe, fluids per the renal diet. Not safe: milk of magnesia, magnesium citrate, Epsom salt, and the Fleet (sodium phosphate) enema (phosphate is dangerous in kidney failure).

Why (tied to the chart): KUB 0345: moderate to large stool, non-obstructive.

Look for: Two safe, two unsafe, an enema named.

Common misses: Listing Fleet as safe because it is "not magnesium".

6B1 Three levels in a row

Copy the three most recent magnesium results and the other new numbers.

Test	Tue 1800	Tue 2100	Wed 0100
Magnesium	4.6	5.6	6.4
Potassium	3.9	4.3	4.5
Creatinine	3.0	3.4	3.5
Calcium	7.7	7.9	8.1
Ionized calcium	1.02	1.06	1.08

Model answer: Mg 4.6, 5.6, 6.4: three steps up. Calcium creeping up toward normal.

Why (tied to the chart): Pattern recognition across three data points.

Look for: Correct Mg trio.

Common misses: Using the 0500 level (not resulted yet).

6B2 What do the last three tell you?

Write what the 4.6, 5.6, 6.4 pattern is called and what it means for the next few hours.

Model answer: A rebound: magnesium is climbing back after dialysis because stored magnesium (cells, bone, gut) is moving into the blood and the kidneys still cannot clear it. It is below the earlier symptomatic range, but rising: recheck at 0500, watch RR, reflexes and rhythm, and plan for repeat dialysis if it goes over 7 or breathing/reflexes worsen (nephrology plan).

Why (tied to the chart): Hospitalist note 0320; nephrology phone plan 0310.

Look for: Uses the word rebound (or describes it) + next steps.

Common misses: Treating it as a lab error.

6B3 Constipation clue

How can constipation keep magnesium high? And the ECG at 0230 shows PR 236 ms (1830 was 204): longer or shorter, and what does that suggest?

Model answer: Magnesium left in the bowel keeps being absorbed into the blood. PR is longer again (236 vs 204 ms): the conduction slowing is coming back with the higher magnesium (strip B pattern).

Why (tied to the chart): ECG 0230: HR 54, PR 236, QRS 114, QTc 478.

Look for: Both parts.

Common misses: Missing the ECG link.

6M Mg-o-meter

New level (Wed 0100): _____ mg/dL Zone: _____

Model answer: X at 6.4 (0100) in the FLUSHING band (5-7), close to 7.

Why (tied to the chart): Third rise in a row.

Look for: Means: drowsy, weaker reflexes possible; reassess often; repeat dialysis threshold is over 7.

Common misses: Plotting 5.6.

6T1 Top 3 priorities

Top 3 priorities, in order.

Model answer: 1) Do not give the milk of magnesia; get the bowel protocol discontinued and a magnesium-free bowel plan (bisacodyl, PEG, KUB; no Fleet). 2) Watch breathing, reflexes and rhythm hourly until the 0500 level; know the call numbers; calcium ready only if symptomatic or QTc over 500. 3) Call the provider about the rising magnesium and abdomen with SBAR; document the stool.

Why: The rebound threatens breathing again; the hidden accomplice is the bowel protocol.

Look for: Accept #1 and #2 swapped if explained.

Common misses: Giving the pink milk; missing the call.

6T2 One question for the provider

The ONE question for the night hospitalist / nephrology:

Model answer: Examples: "What magnesium level or finding means repeat dialysis?" "Can you discontinue the bowel protocol and order bisacodyl and a KUB?" "How often should I check reflexes and RR until the 0500 level?"

Look for: Specific, with a number or an order.

Common misses: Asking about discharge.

6T3 SBAR

SBAR call to the night hospitalist about rebound magnesium and constipation.

Model answer: S: "The 0100 magnesium is 6.4, up from 4.6 after dialysis and 5.6 at 2100. HR 56, BP 104/60, RR 12, reflexes 1+, drowsy." B: "74-year-old, dialysis yesterday for Mg 10.9; no stool in 4 days; abdomen distended with hypoactive bowel sounds. The bowel protocol ends in milk of magnesia and he is asking for it; I have not given it." A: "Rebound magnesium with retained gut magnesium." R: "Please discontinue the bowel protocol, order bisacodyl and a KUB, call nephrology about repeat dialysis if it goes over 7, and tell me when to call you again."

Why: Matches model SBAR #6.

Look for: Mentions that milk of magnesia was NOT given.

Common misses: No clear request for the protocol to be stopped.

6P Predict the plot twist (bonus)

No wrong answers. Give the bonus credit for any thoughtful guess with a reason. What actually happens next: see the next checkpoint. Clue card: **CLUE UNLOCKED #6**: The accomplice was a standing order. Teams that catch hidden magnesium in a protocol prevent a second crisis. Question any order or protocol that contains magnesium in a kidney patient.

CASE CLOSED: KEY — Wed 0700

Final check (Wed 0700)

Set the monitor to	Value
Temp (F)	98.0
HR	68
BP (MAP)	124/70 (88)
RR	16
SpO2 / O2	96% room air
EtCO2	not shown
Pain	2/10 back
Rhythm	Sinus rhythm 68, PR 0.20 s
POC glucose	112

Check	Expected
Level of consciousness	Awake, alert, oriented x4, CAM negative, full sentences
Breathing	RR 16
Reflexes and grip	Reflexes 2+ (normal), grip strong
Lungs, neck veins, ankles	Fine crackles at both bases, trace ankle edema, neck veins mildly up
Abdomen and stool	Soft, bowel sounds active, passing gas; large stool at 0345
Urine and weight	Last hour 85 mL (0300-0700 hourly 55, 65, 75, 85 = 280 mL); weight 90.0 kg (88.0 on arrival)

Test	Worst	Wed 0500 (final)
Magnesium	10.9 (Tue 1315)	5.3
Potassium	5.8 (Tue 1315)	4.3
Creatinine	5.3 (Tue 1315)	3.3

Whole-story Mg-o-meter (all draws, mg/dL): Oct 20 2.4; Tue 0440 7.8; 0900 8.7; 1115 9.5; 1315 10.9; 1700 7.2; 1800 4.6; 2100 5.6; Wed 0100 6.4; 0500 5.3. Arrows: calcium gluconate first given in the ED at 0555 (before the 0440 draw result), dialysis 1500-1800, hidden magnesium caught at 0300 (bowel protocol) and in the 0700 bag.

Timeline model (page 2 of the closing)

Stop	Mg	Big problem	What a good squad did
Tue 0700	7.8	Arrival: drowsy, slow heart rate, reflexes 1+; the bag of bottles	Checked ABCs, reflexes and rhythm; asked about OTC products; stopped all magnesium sources; safe fluids
Tue 1100	8.7	Reflexes gone, RR 10, HR 44, BP 84/48	Called the rapid response; calcium gluconate on the monitor; oxygen but watched RR; SBAR
Tue 1500	10.9	Peak Mg, breathing depression, wide QRS, emergency dialysis decided	Dialysis checklist (consent, catheter rules, x-ray); ICU thresholds; calcium 2 g; BP/HR q15
Tue 1900	4.6	After dialysis: Mg 4.6 but glucose 54 and low calcium	Rule of 15 with swallow screen (apple juice); called for D10W; post-dialysis checks
Tue 2300	5.6	Mg climbing again (rebound) and new delirium	Non-drug measures, protected lines, did not give diphenhydramine, asked for melatonin
Wed 0300	6.4	Rebound, constipation, Ray asks for "the pink milk"	Did NOT give milk of magnesia; got the protocol stopped; bisacodyl, KUB; watched breathing and reflexes
Wed 0700	5.3	Recovery: alert, reflexes 2+, urine 85 mL/h, a large stool at 0345	Teaching with teach-back; Foley removal plan; handoff

Scoring: Full credit = all seven rows have a Mg value and a specific event and action in the squad's own words.

Common misses: Copying only times; leaving out why the level changed.

CT Teach-back to Eleanor

Eleanor says: "What do I do with all those bottles at home? He gets so constipated from his back medicine, and the heartburn medicine always helped."

Model answer: Three plain-language tips, for example: (1) "Do not give Ray milk of magnesia, Maalox or Mylanta-type antacids, magnesium pills or Epsom salts. Look for the word MAGNESIUM on the label. His kidneys cannot get rid of it, and it made him very sleepy and slowed his heart." (2) "For constipation, heartburn, cramps or pain, call his kidney doctor or the pharmacist FIRST. Ask about MiraLAX or senna for constipation and Tylenol (acetaminophen) instead of ibuprofen." (3) "Bring every bottle, even the store-bought ones, to every visit, and say 'Ray has kidney disease.' If he is vomiting or not drinking, call before giving his blood-pressure pill or water pill. Call us or 911 if he is very sleepy, breathing slowly, weak, fainting or making very little urine."

Why (tied to the chart): Based on INSTRUCTOR_GUIDE.md section 8.5 and the 0700 teaching points (no magnesium, no NSAIDs, label reading, safe alternatives, sick-day rule, bring all bottles, warning signs).

Look for: Accept any three correct tips in plain language (no jargon like "hypermagnesemia"), with at least one named safe alternative. A great teach-back question: "Show me which bottle on the shelf you would not buy."

Common misses: Jargon; "avoid antacids" with no alternatives; no mention of reading labels; forgetting that Ray also took ibuprofen.

Reflection strips and peer check: scored individually (1 point each, see score sheet). Any honest, specific reflection earns the point: "one thing I did well, one thing I will do differently, one question".

RUBRIC SCORING GUIDE: 20 POINTS

generous partial credit

Category	Pts	FULL	GOOD (partial)	GETTING THERE
Teamwork and participation	5	5 points. All names on the cover. Roles rotated every checkpoint and initialed. Every teammate clearly contributed in the huddles. Every teammate finished the peer check.	3 to 4 points. Roles rotated most of the time (a few initials missing). Most teammates contributed. Peer check done by most.	1 to 2 points. One person did most of the work, or roles were not rotated. Peer check missing. (0 = nothing handed in.)
Clinical reasoning at each checkpoint	6	6 points. All six checkpoints have thoughtful answers in the Detective, Interpreter and Bedside boxes, using real chart numbers and giving reasons.	4 to 5 points. Most checkpoints are thoughtful. One or two are thin (answers without numbers or reasons).	1 to 3 points. Many boxes are blank or one-word answers, or the chart facts are mostly missing.
Safety, priorities and SBAR communication	5	5 points. At each of the five calls (1100, 1500, 1900, 2300, 0300): safe priority order, a specific question, and a complete SBAR with numbers and a clear request.	3 to 4 points. Mostly safe priorities and SBARs. A part is missing sometimes (no numbers, no request).	1 to 2 points. Priorities are unsafe or very vague, or SBARs are mostly empty.
Completeness and reflection	4	4 points. Every checkpoint page is finished (Mg-o-meter marked, vitals logged, prediction written). Case Closed timeline done. Teach-back to Eleanor done. Individual reflection done.	2 to 3 points. One or two of these four parts are partly done.	0 to 1 point. Most of the end pages or checkpoint pages are blank.

Examples of full and partial credit

Teamwork and participation (5). FULL (5): three names on the cover, initials A/B/C change at every checkpoint, the huddle boxes show three different handwritings or "who said it" notes, three peer checks.

PARTIAL (3): roles rotated at four of six checkpoints, one student mostly writing. Peer checks done. (2): everyone's name on the cover but the same initials on every strip.

Clinical reasoning (6). FULL (1 point at a checkpoint): "Mg went 7.8 to 8.7 even with 1 L of saline and furosemide because his kidneys only made 30-55 mL/h (creatinine 4.9 to 4.8, FENa 0.6%)."

HALF (0.5): "His magnesium is going up because of kidney problems" (right idea, no numbers). ZERO: blank or "I do not know" in most boxes of that checkpoint.

Safety, priorities and SBAR (5). FULL (1 point at a checkpoint): priorities start with airway/breathing or safety, a specific question ("Do you want calcium gluconate 1 g now?"), SBAR with RR 10, HR 44, BP 84/48 and "I need you at the bedside now".

HALF (0.5): priorities fine but SBAR missing the Recommendation, or SBAR with no numbers. ZERO: priorities start with paperwork and no SBAR.

Completeness and reflection (4). FULL (4): all checkpoint pages finished (Mg-o-meter marked each time), timeline filled in with all seven Mg values, three plain-language tips, reflection written.

PARTIAL (2 to 3): Mg-o-meter marked only some times, or timeline half done, or reflection one line. (0 to 1): last three pages blank.

Bonus: +1 for all six "Predict the plot twist" boxes completed with a thoughtful guess (no wrong answers). Default cap: total 20 (decide if you want 21).

GROUP SCORE SHEET

one per squad

Squad name: _____ Members: _____ Date: _____

Checkpoint	Clinical reasoning (1 each, 6)	Safety, priorities, SBAR (1 each, 5)	Roles initialed (all six = 2)	Huddle shows whole team (all six = 2)	Notes
Tue 0700	0 0.5 1	warm-up	☐	☐	
Tue 1100	0 0.5 1	0 0.5 1	☐	☐	
Tue 1500	0 0.5 1	0 0.5 1	☐	☐	
Tue 1900	0 0.5 1	0 0.5 1	☐	☐	
Tue 2300	0 0.5 1	0 0.5 1	☐	☐	
Wed 0300	0 0.5 1	0 0.5 1	☐	☐	

Group-scored item	Max	Squad score
Clinical reasoning (sum of six checkpoints)	6	
Safety, priorities, SBAR (sum of 1100, 1500, 1900, 2300, 0300)	5	
Teamwork: roles rotated and initialed at every checkpoint	2	
Teamwork: huddles show the whole team	2	
Completeness: all checkpoint pages finished (Mg-o-meter, vitals, prediction)	1	
Completeness: Case Closed timeline	1	
Completeness: teach-back to Eleanor	1	
GROUP SUBTOTAL	18	

Individual score line (copy the group subtotal, add the two individual points)

Student name	Group subtotal (18)	Peer check done (1)	Reflection done (1)	Bonus (+1)	TOTAL (20)

Peer checks are for the instructor only. If one student's peer ratings are all "sometimes", talk to the squad before lowering that student's Teamwork.

DEBRIEF HUDDLE: 15 MINUTES

questions from INSTRUCTOR_GUIDE.md section 8

Opening (2 min): Reactions: "One word for how that felt?" Every student answers. Ground rules: "We talk about what we did, not who we are."

Q1 (guide 8.2 #1): Why was Ray at such high risk for magnesium toxicity? (Failing kidneys are the only exit; three magnesium products for weeks; dehydration; kidney-hurting medicines; opioid constipation.)

Q4 (#4): Which bedside finding told you the magnesium was getting worse before the breathing failed? (Reflexes 1+ to 0; more sleepiness; low BP; slower HR; longer PR.)

Q5 (#5): Why does calcium gluconate work, and why is it not enough? (Opposes magnesium at nerve/muscle and heart for 30-60 minutes; does not lower the level; dialysis removes it.)

Q12 (#12): What is rebound and what did you do about it? (4.6, 5.6, 6.4; stored magnesium and gut magnesium; repeat levels, watch RR/reflexes, clear the bowel without magnesium.)

Q14 (#14): What was the hidden hazard at 0300? (The bowel protocol ended in milk of magnesia; question any order or protocol with magnesium in a kidney patient.)

Q15 (#15): What will you teach Ray and Eleanor and how will you know they understood? (No magnesium or NSAIDs, label reading, safe alternatives, sick-day rule, bring all bottles, warning signs; teach-back.)

Close (1 min): Each squad says its top clue (the box on closing page 2). Take-aways: ask about OTC products; reflexes and RR before SpO2; calcium protects, dialysis removes; read every order for magnesium; call early with SBAR.

If you have more time, use the full PEARLS structure and the "what went well / improve" table in INSTRUCTOR_GUIDE.md section 8.